INTERN ID : CITS3033

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NO.OF.WEEKS : 4 WEEKS

ENDPOINT DETECTION & RESPONSE (EDR) BUILD

1. INTRODUCTION

Endpoint Detection and Response (EDR) is a cybersecurity solution that continuously monitors endpoint devices for suspicious activities and security threats.

This project implements a simplified EDR system using Python to demonstrate the basic workflow of threat monitoring, detection, logging, and visualization.

2. OBJECTIVE

The primary objective of this project is to:

- Monitor endpoint activities.
- Detect suspicious processes.
- Monitor file system events.
- Record security incidents.
- Visualize detected threats through a dashboard.

3. PROBLEM STATEMENT

Organizations face various endpoint security threats such as malware execution, unauthorized file modifications, and suspicious process activity.

Traditional antivirus solutions may not provide sufficient visibility into endpoint behavior.

This project addresses this issue by creating a lightweight monitoring solution capable of detecting and logging suspicious activities.

4. SYSTEM ARCHITECTURE

Endpoint Activities

↓

Process Monitoring Module

↓

File Monitoring Module

↓

Detection Engine

↓

SQLite Database

↓

Streamlit Dashboard

5. MODULES

5.1 PROCESS MONITORING MODULE

Purpose:
Monitor running processes on the endpoint.

Functions:

- Enumerate active processes.
- Compare processes against blacklist.
- Generate alerts for suspicious processes.

Technology:
Psutil Library

5.2 FILE MONITORING MODULE

Purpose:
Track changes in monitored directories.

Functions:

- Detect file creation.
- Detect file modification.
- Detect file deletion.

Technology:
Watchdog Library

5.3 DETECTION ENGINE

Purpose:
Identify potentially malicious activities.

Functions:

- Rule-based threat detection.
- Alert generation.
- Severity assignment.

Severity Levels:

- Low
- Medium
- High
- Critical

5.4 DATABASE MODULE

Purpose:
Store detected security events.

Technology:
SQLite

Database Table:

alerts

Fields:

- id
- time
- process
- severity

5.5 DASHBOARD MODULE

Purpose:
Provide visual representation of detected threats.

Technology:
Streamlit(Streamlit)

Features:

- Alert Table
- Threat Count
- Severity Statistics
- Threat Visualization Charts

6. WORKFLOW

1. Endpoint activities occur.
2. Monitoring modules capture events.
3. Detection engine evaluates events.
4. Alerts are generated.
5. Alerts are stored in SQLite.
6. Dashboard retrieves alert data.
7. User views detected threats.

7. TESTING

PROCESS DETECTION TEST

Input:
Launch a blacklisted process.

Expected Result :

Threat alert generated.

FILE MONITORING TEST

Input:
Create a file in monitored directory.

Expected Result:

File event logged.

DASHBOARD TEST

Input:
Insert sample alerts.

Expected Result:

Dashboard displays threat information.

8. RESULTS

The system successfully:

- Monitored endpoint processes.
- Tracked file system activities.
- Logged security events.
- Displayed threat information through a dashboard.

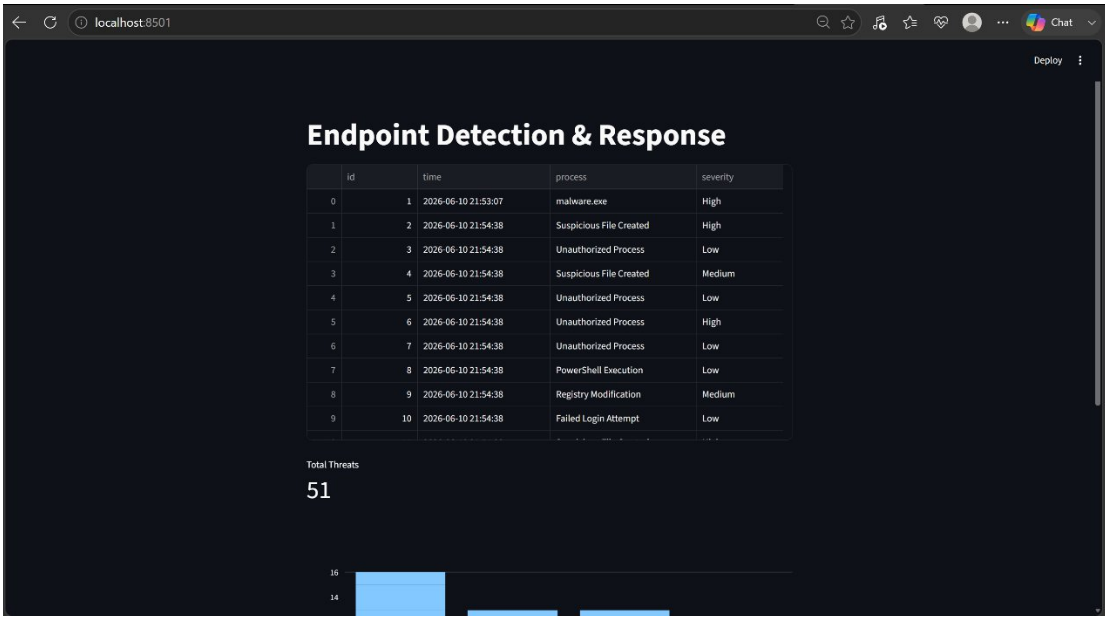
9. LIMITATIONS

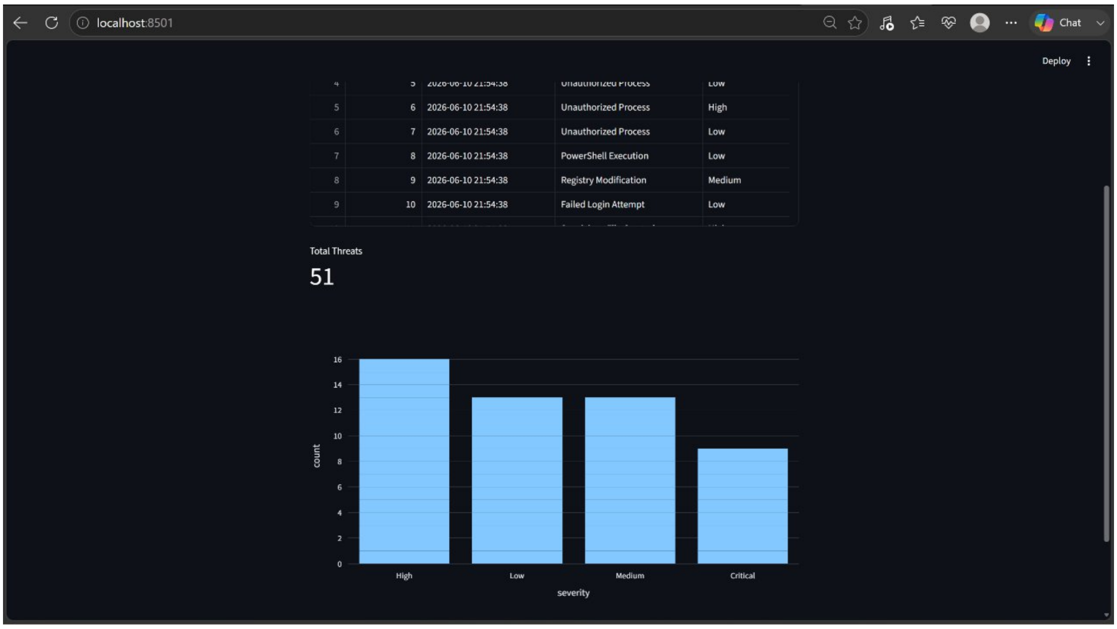
- Rule-based detection only.
- No machine learning integration.
- Limited network monitoring.
- Designed for educational purposes.

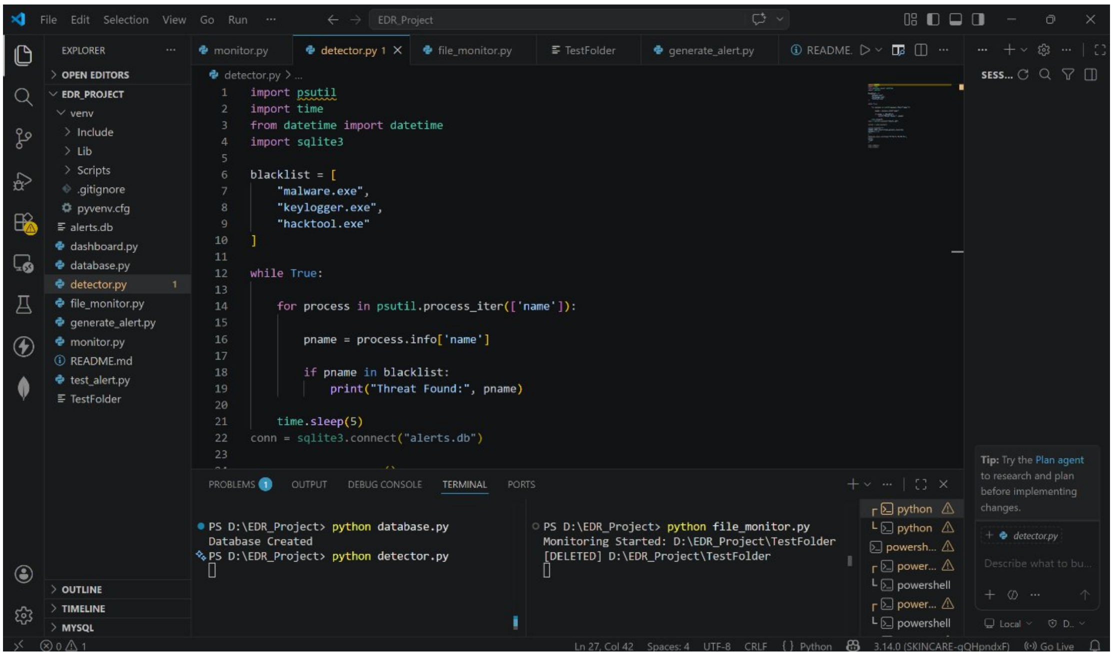
10. CONCLUSION

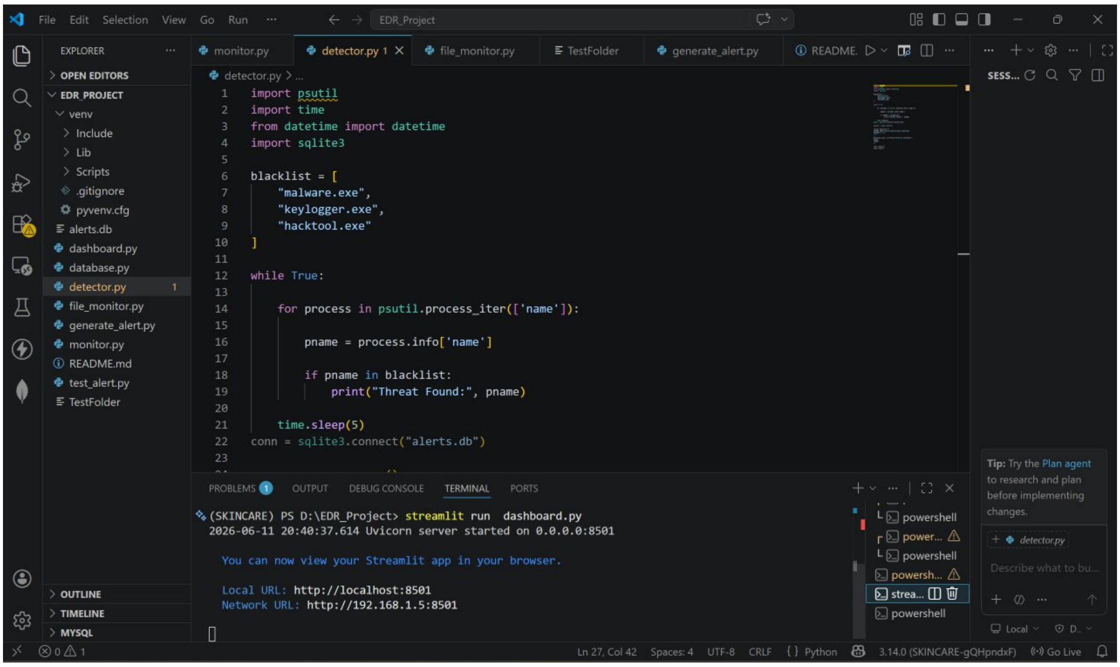
The Endpoint Detection & Response Build project demonstrates the fundamental concepts of endpoint security by integrating monitoring, detection, logging, and visualization components into a single cybersecurity solution. The project provides practical exposure to security monitoring techniques and serves as a foundation for advanced EDR development.

OUTPUT IMAGES






EDR_Project
monitor.py
detector.py
file_monitor.py
TestFolder
generate_alert.py
README
EXPLORER
OPEN EDITORS
EDR_PROJECT
venv
Include
Lib
Scripts
.gitignore
pyvenv.cfg
alerts.db
dashboard.py
database.py
detector.py
file_monitor.py
generate_alert.py
monitor.py
README.md
test_alert.py
TestFolder
import psutil
import time
from datetime import datetime
import sqlite3
blacklist = [
"malware.exe",
"keylogger.exe",
"hacktool.exe"
]
while True:
for process in psutil.process_iter(['name']):
pname = process.info['name']
if pname in blacklist:
print("Threat Found:", pname)
time.sleep(5)
conn = sqlite3.connect("alerts.db")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS D:\EDR_Project> python database.py
Database Created
PS D:\EDR_Project> python detector.py
PS D:\EDR_Project> python file_monitor.py
Monitoring Started: D:\EDR_Project\TestFolder
[DELETED] D:\EDR_Project\TestFolder
Tip: Try the Plan agent to research and plan before implementing changes.
OUTLINE
TIMELINE
MYSQL
Ln 27, Col 42
Spaces: 4
UTF-8
CRLF
Python
3.14.0 (SKINCARE-gQHpndxF)
Go Live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TERMINAL
(SKINCARE) PS D:\EDR_Project> streamlit run dashboard.py
2026-06-11 20:40:37.614 Uvicorn server started on 0.0.0.0:8501
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://192.168.1.5:8501
Tip: Try the Plan agent to research and plan before implementing changes.
Ln 27, Col 42
Spaces: 4
UTF-8
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